Use Cases

Use Cases

This page helps you learn ReconX by scenarios. Each use case includes:

- What business problem is being solved
- Which datasets are typically compared

- Starter rule ideas (exact rules first, then tolerance rules)
- What to do in the workspace (reconcile, review, link with notes, send to ResolveX)

TIP

If you are new to ReconX, start with **Example 1** and follow the same operating steps in Workspace.

Example 1: Daily Bank Reconciliation

Business Context

Finance teams in companies of all sizes face the daily challenge of reconciling bank statements against accounting records. Without automation:

- **Manual effort**: 4-8 hours daily using spreadsheets and VLOOKUP formulas
- **Error-prone**: Copy-paste mistakes and formula errors lead to mismatches
- **No audit trail**: Excel files passed via email with lost context
- **Delayed detection**: Fraud or missing transactions discovered weeks later

Banks and financial institutions handle thousands of transactions daily, making manual reconciliation impractical and risky. Regulatory compliance (SOX, Basel) requires complete audit trails and timely reconciliation.

Solution Overview

ReconX automates bank reconciliation by:

- **Automated matching**: Compares bank statements against ledger in seconds (vs hours manually)
- **Rule-based intelligence**: Exact match on reference numbers + tolerance on dates/amounts
- **AI optimization**: Learns patterns and suggests improved rules automatically
- **Exception routing**: Unmatched items flow to ResolveX for investigation with audit trail

Typical results: 85-95% auto-match rate, reducing manual effort from 4-8 hours to 30 minutes.

Setup Required

You'll need to configure:

- **Ops Process**: Link bank statement Data Set to ledger Data Set

  Example: `Daily Bank Reconciliation` connecting "HDFC Bank Statements" to "SAP General Ledger"

- **Data Sets**: Define column mappings for both sources

Bank: `Date`, `Description`, `Debit`, `Credit`, `Balance`, `Reference` Ledger: `Posting Date`, `Narration`, `Amount`, `GL Code`, `Document Number`

- **Recon Rules**: Matching criteria (can be auto-generated via Optimizer)

  Rule 1: `Amount` (NUMERIC) with `₹1` tolerance Rule 2: `Date ↔ Posting Date` (DATE) with `2 days` tolerance Rule 3: `Reference ↔ Document Number` (STRING) exact match

See Ops Process Configuration for detailed setup instructions.

Operating Steps

1. **Select Ops Process and Period**

   Navigate to Dashboard > ReconX > Workspace, select "Daily Bank Reconciliation" and period (e.g., `P-30D` for last 30 days).

2. **Upload bank statement and ledger files** (CSV format)

   Click upload for Data Source 1 (Bank) and Data Source 2 (Ledger). System validates headers and ingests records.

   | TIP

   Use DataX for automated ingestion if files arrive via email/SFTP daily.

3. **Run Recon Optimizer** (first time or if rules need improvement)

   Click **Recon Optimizer** → Wait for analysis → Review suggested rules in **View & Edit Recon Rule**

4. **Execute reconciliation**

   Click **Reconcile** button. System processes all records and updates status to **Reconciled** or **Open**.

   Expected result: ~85% auto-matched on first run (95%+ after optimization)

5. **Review unmatched items** (the remaining 5-15%)

   Filter table by status **Open** → Double-click records to investigate:

   - **Valid match found**: Use **Suggested Matches** → Select match → Click **Link** → Add notes explaining why (audit trail)
   - **No match exists**: Use **Send to ResolveX** → Select workflow → Add notes → System routes to exception management
   - **Need more data**: Add notes and revisit next cycle

6. **Download reconciliation report**

   Click download icon → Select format → Save status report for audit and management review

7. **Handle exceptions in ResolveX** (if workflow enabled)

Navigate to Dashboard > ResolveX → Process items through approval/investigation queues

Full workspace operations guide: Workspace

Expected Business Outcomes

After completing this workflow daily, you should see:

- **85-95% auto-match rate** within first week

  Verify by checking reconciliation percentage in workspace summary

- **Manual effort reduced from 4-8 hours to 30 minutes**

  Time saved on VLOOKUP formulas, copy-paste, email coordination

- **Same-day exception detection** instead of month-end surprises

  Open items visible immediately with assignment to resolution teams

- **Complete audit trail** for compliance

  All links, delinks, notes, and status changes timestamped with user attribution

- **Fraud detection within 24 hours**

  Missing transactions or amount discrepancies flagged for immediate investigation

Related Scenarios

- Example 2: Payment Gateway Reconciliation - Similar pattern with multi-match complexity
- How Rules Optimize Reconciliation - Deep dive into rule tuning

Example 2: Payment Gateway Reconciliation

Business Context

E-commerce companies and payment processors handle thousands of online transactions daily across multiple payment gateways (Stripe, PayPal, Razorpay). Challenges include:

- **Split payments**: Single order paid via multiple partial payments (e.g., ₹1000 order split into ₹500 + ₹500)
- **Gateway fees**: Settlement amount differs from order amount due to deducted fees
- **Refunds and chargebacks**: Negative amounts or reversal entries complicate matching
- **Delayed settlements**: Payment received days after order due to bank processing
- **Multiple gateways**: Each gateway has different field names and settlement formats

Without automated reconciliation, finance teams spend hours manually matching orders to settlements, leading to revenue leakage and delayed fraud detection.

Solution Overview

ReconX handles payment gateway complexity by:

- **Multi-match support**: One order matches multiple partial payments (many-to-1) or one payment covers multiple orders (1-to-many)
- **Tolerance handling**: Amount matching accounts for gateway fees with percentage/absolute tolerance
- **Status-aware matching**: Considers transaction status (captured, refunded, cancelled) in reconciliation logic
- **Cross-gateway reconciliation**: Normalize different gateway formats for unified matching

Result: 80-90% auto-match rate even with complex payment scenarios.

Setup Required

You'll need to configure:

- **Ops Process with Multi-Match enabled**

  Example: `Payment Gateway Reconciliation` with "Requires Multi-Match" = Yes, "One-to-Many" = Yes

- **Data Sets for order system and gateway settlements**

  Orders: `Order ID`, `Order Date`, `Amount`, `Payment Method`, `Status` Gateway: `Merchant Order ID`, `Transaction ID`, `Settlement Amount`, `Gateway Fee`, `Settlement Date`

- **Recon Rules accounting for fees and timing**

  Rule 1: `Order ID` ↔ `Merchant Order ID` (STRING) exact match Rule 2: `Amount` ↔ `Settlement Amount` (NUMERIC) with `2%` tolerance (for fees) Rule 3: `Order Date` ↔ `Settlement Date` (DATE) with `3 days` tolerance

See Multi-Match Configuration for setup details.

Operating Steps

1. **Select Payment Gateway Ops Process**

   Navigate to Workspace → Select "Payment Gateway Reconciliation" → Choose period

2. **Upload order export and gateway settlement report**

Upload orders CSV from your e-commerce system and settlement report from payment gateway (Stripe/PayPal CSV export)

3. **Enable Multi-Match in rules** (if not already)

   Open **View & Edit Recon Rule** → Mark one rule column as "Multi Match" (typically Amount) → Save

   TIP

   Only mark ONE column for multi-match to avoid ambiguous aggregation

4. **Run reconciliation with multi-match**

   Click **Reconcile** → System identifies 1-to-1 matches AND aggregates split payments

   Expected scenarios:

   - **Normal match**: Order ₹500 ↔ Payment ₹500 (auto-matched)
   - **Split payment**: Order ₹1000 ↔ Payment ₹500 + ₹500 (multi-match linked)
   - **Gateway fee**: Order ₹1000 ↔ Settlement ₹980 (matched with 2% tolerance)
   - **Missing payment**: Order ₹750 ↔ No payment found (stays Open)

5. **Review Open items and classify exceptions**

   Double-click unmatched orders → Investigate:

   - **Refund/chargeback**: Check for negative amounts or reversal entries → Link manually with notes
   - **Delayed settlement**: Not yet settled by bank → Leave Open and revisit next day
   - **Fraudulent order**: No payment found and customer dispute → Send to ResolveX for investigation
   - **Fee mismatch**: Actual fee > expected tolerance → Adjust rule or link manually

6. **Send persistent exceptions to ResolveX**

   Select items with no resolution path → **Send to ResolveX** → Choose workflow (e.g., "Payment Disputes") → Add notes

   ResolveX queues: - "Pending Gateway Response" → awaiting gateway support reply - "Fraudulent Transactions" → escalated to fraud team - "Write-Off Approval" → small amounts pending manager approval

7. **Download reconciliation report for finance**

   Export reconciled/open/exception summary for revenue recognition and audit

Full workspace operations guide: Workspace

Expected Business Outcomes

After implementing payment gateway reconciliation:

- **80-90% auto-match rate** including split payments

  Multi-match handles most partial payment scenarios automatically
- **Revenue leakage reduced to <0.1%**

  Missing settlements detected within 24 hours instead of month-end
- **Chargeback detection within same day**

  Negative amounts flagged immediately for fraud investigation
- **Gateway fee reconciliation automated**

  Tolerance rules eliminate manual fee calculation
- **Cross-gateway visibility**

  Single dashboard shows reconciliation status across all payment providers

Related Scenarios

- Example 3: N-Way Settlement Reconciliation - Extends to 3+ data sources
- Common Reconciliation Patterns - See Pattern 6 for split payment details

Example 3: N-Way Settlement Reconciliation

Business Context

Payment processors, banks, and large enterprises need to reconcile across **three or more data sources** simultaneously:

1. **Internal transaction log** - System of record for all transactions
2. **Bank settlement reports** - Actual funds received/disbursed
3. **Merchant/partner reports** - Third-party confirmation of settlements

Challenges unique to N-way: * **Partial matches**: 2 out of 3 sources match, but third is missing * **Ownership confusion**: Which team investigates when bank record is missing vs merchant record? * **Volume complexity**: 10,000+ transactions daily across multiple settlement cycles * **Regulatory pressure**: Financial institutions must reconcile within T+1 (next business day)

Without N-way reconciliation, teams maintain separate 2-way reconciliations and manually correlate results, leading to gaps and delays.

Solution Overview

ReconX N-Way Reconciliation handles 3+ sources by:

- **Unified matching**: Single reconciliation process across all sources (not separate 2-way reconciliations)
- **Partial match detection**: Identifies when 2 sources agree but third disagrees or is missing
- **Ownership routing**: Configurable exception routing based on which source has gaps
- **Settlement cycle awareness**: Groups transactions by settlement batch/cycle for aggregate matching

Result: Single dashboard showing full 3-way match status with clear exception ownership.

Setup Required

You'll need to configure:

- **N-Way Recon Process**: Group multiple Ops Processes for unified N-way matching

  Example: `Payment Processor Settlement` containing: - Ops Process 1: Internal vs Bank - Ops Process 2: Internal vs Merchant - Ops Process 3: Bank vs Merchant

- **Individual Ops Processes** for each 2-way comparison (prerequisites)

  Each Ops Process defines Data Sets and rules as normal 2-way reconciliation

- **Exception routing workflows** for partial matches

  Bank missing → Treasury/Banking team workflow Merchant missing → Merchant Operations team workflow Internal missing → Engineering/Operations team workflow

See N-Way Recon Process for setup details.

Operating Steps

1. **Select N-Way Recon Process**

   Navigate to Dashboard > ReconX > N-Way Workspace → Select "Payment Processor Settlement"

2. **Verify all sources ingested**

   Check that data appears for all three sources (Internal, Bank, Merchant). If missing, upload or trigger ingestion.

3. **Run N-way reconciliation**

   Click **Reconcile** → System executes all 2-way comparisons and aggregates results

Match states displayed:

Internal	Bank	Merchant	Status & Action
✓	✓	✓	**Fully Reconciled** - No action needed
✓	✓	✗	**Merchant discrepancy** - Route to Merchant Ops
✓	✗	✓	**Bank discrepancy** - Route to Treasury
✓	✗	✗	**Critical issue** - Escalate immediately
✗	✓	✓	**Internal missing** - Engineering investigation

4. **Filter by partial match status**

   Use N-Way workspace filters to view: - Fully reconciled (all 3 match) → No action - 2-of-3 matched → Investigate missing source - 1-of-3 or none → Critical exceptions

5. **Route partial matches to appropriate teams**

   Based on which source is missing:

   - **Bank missing (Internal+Merchant match)**: Select records → **Send to ResolveX** → "Bank Settlement Issues" workflow → Assign to Treasury
   - **Merchant missing (Internal+Bank match)**: Send to "Merchant Reconciliation" workflow → Assign to Merchant Operations
   - **Internal missing (Bank+Merchant match)**: Send to "System Integrity" workflow → Assign to Engineering (potential data loss)

6. **Escalate critical (0-of-3 or 1-of-3) exceptions**

   Items not matching any source require immediate attention → Send to "Critical Exceptions" workflow with high priority

7. **Download N-way status report**

   Export showing reconciliation percentage per source and outstanding exception counts by ownership

Full N-way operations guide: N-Way Workspace

Expected Business Outcomes

After implementing N-way reconciliation:

- **Single source of truth** for multi-party settlement status

  No more reconciling separate 2-way reconciliations manually

- **Clear exception ownership** from day one

  Partial match routing eliminates "who should handle this?" delays

- **T+1 regulatory compliance** achieved

  Automated daily reconciliation meets next-business-day requirements

- **Reduced settlement delays** by 60-70%

  Faster identification of missing records enables same-day resolution

- **Fraud pattern detection across sources**

  Discrepancies between internal and external sources flag potential fraud

Related Scenarios

- Example 1: Bank Reconciliation - Foundation 2-way pattern
- Example 2: Payment Gateway - Multi-match complexity applicable to N-way

TIP

For operational troubleshooting (no matches, too many matches, cannot link), use Troubleshooting.

Common Reconciliation Patterns

These patterns appear in most reconciliations. Use them as a checklist when building rules.

Pattern 1: Normal match (1-to-1)

Goal: One record in data source 1 matches one record in data source 2.

Recommended rules (starter set):

- Strong identifier (exact): `Transaction ID / Reference / Document Number`
- Supporting rule (often exact): `Currency` , `Account` , `GL Code`
- Supporting rule (often tolerance): `Amount` , `Date`

Pattern 2: Timing differences (date mismatch)

Goal: Match records where posting date differs by a small number of days.

Recommended approach:

- Keep the identifier exact if possible
- Use date tolerance (e.g., `±2 days`) only as a supporting rule

Pattern 3: Fees, charges, and rounding

Goal: A payment amount differs slightly due to fees or rounding.

Recommended approach:

- Match on identifier + net amount with a small tolerance
- If fees are separate lines, treat as a multi-record scenario and investigate multi-match

Pattern 4: Refunds, chargebacks, and reversals

Goal: A transaction is reversed later, or a negative amount appears.

Recommended approach:

- Include `Transaction Type` / `DrCr` / sign checks where available
- Use ResolveX workflow for investigation when reversal logic is business-specific

Pattern 5: Duplicates and repeats

Goal: One side has duplicates due to reprocessing; the other side has single entry.

Recommended approach:

- Add a stronger identifier rule before loosening tolerance
- Use ResolveX if duplicates require approval or root-cause classification

Pattern 6: Split payments / split settlements (many-to-1 or 1-to-many)

Goal: A single record on one side matches multiple records on the other side.

Recommended approach:

- Enable Multi-Match only when your Ops Process is designed for aggregation.
- In rule setup, mark only one rule column as `Multi Match` (`colForMultiMatch`) to avoid ambiguous aggregation.

Pattern 7: Missing item (no match)

Goal: There is no corresponding record.

Recommended approach:

- Do not force-match by increasing tolerance too much
- Send to ResolveX for investigation/closure when needed

Mini Example: Exact + Tolerance Rules (Starter Template)

Use this when you need a safe baseline.

- Rule 1 (Exact): `Transaction ID` must match
- Rule 2 (Tolerance): `Amount` matches within `₹10` (`ABSOLUTE`) or within `0.5%` (`PERCENTAGE`)

Expected behavior:

- When Rule 1 matches and Rule 2 is within tolerance, records can be linked and become **Reconciled**.
- If Rule 1 is missing/blank, do not rely only on amount tolerance (it will cause false matches).
